

Integer Linear Forms for a Factorial Reciprocal Series

An integral basis and a finite gcd formula

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ABSTRACT

We classify the integer vectors that cancel prescribed initial weighted sums in linear forms $MS + k$, where $S = \sum_{n \geq 2} (n! - 1)^{-1}$. An integral basis reduces the support restriction to one linear Diophantine equation. The possible values of M are determined by a gcd over fewer than $2D^2$ indices when cancellation is imposed through D . For $D = 4$ we construct a vector with the least positive value $M = 1380$. Nonintegrality for a family with the required denominator divisibility remains open.

1 Introduction

Erdős asked whether replacing $n!$ by $n! - 1$ preserves the irrationality of the reciprocal sum [1, p. 102]. The question appears as Problem 68 in Bloom’s catalogue [2].

Problem 1.1 (Erdős Problem #68). Is

$$S = \sum_{n \geq 2} \frac{1}{n! - 1}$$

irrational?

The elementary proof that e is irrational uses two properties of factorials: they clear every earlier denominator, and they leave a small positive tail. The first property fails here. If $H_m = \sum_{n=2}^m (n! - 1)^{-1}$, then already $3!H_3 = 36/5$. We shall instead construct linear forms $MS + k$, with $M, k \in \mathbb{Z}$, in which several initial summands cancel. A non-integral form excludes every rational value $S = a/q$ for which $q \mid M$.

We classify the integer coefficient vectors that cancel the first $D - 1$ weighted sums. Starting with adjacent factorial differences, we correct each vector using those already constructed at its proper divisors. The resulting basis isolates each weighted sum (Theorem 2.1). The triangular change of basis is invertible over $\mathbb{Z}$, so it gives coordinates for every integer solution, not just a way to construct some solutions.

* *Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

The desired vectors are supported on indices at least two. Temporarily allowing index one makes the basis construction possible; removing it then amounts to one linear Diophantine equation. Its solvability determines the possible values of M . The gcd that occurs in this equation is defined by an infinite sequence, but Theorem 3.1 computes it on a specified finite interval.

When cancellation is imposed through $D = 4$, the least positive value of M is 1380. We derive a vector attaining this value and bound its remainder between consecutive integers. The classification itself does not assume that S is rational. An irrationality proof would require, for each $q > 0$, a nonintegral form whose coefficient M is divisible by q . The companion constructs primitive vectors with this divisibility property; nonintegrality of their remainders remains unproved.

Hančl and Tijdeman's tail-integrality lemma treats factorial series with integer coefficients [4, Lemma 2.1 and the following remark, p. 385]. Their factorial-scaled partial sums are integers. Here the scaled partial sum need not be integral. Instead, the weight congruences give $\mathcal{R} - MS \in \mathbb{Z}$ (1); the assumption $S = a/q$ and $q \mid M$ then makes $\mathcal{R}$ integral. Koepf and Schmersau's factorial-digit criterion [3, Example 3.2, p. 121] is used in the companion's digit argument to give an equivalent eventual-pattern condition for $S - e + 2$. Failure of that pattern at arbitrarily large indices also remains unproved.

Sections 2 and 3 give the coordinate construction and finite gcd. Section 4 uses them to construct and test a vector attaining the least positive value of M . Section 5 states the remaining nonintegrality question, with the secondary arithmetic criteria developed in the companion.

2 An integral basis

Since $d! \equiv 1 \pmod{d! - 1}$, dividing $n!$ by an integer power of $d!$ preserves its residue whenever the quotient is integral. We choose the exponent so that the quotient changes predictably at successive indices. For $d \geq 2$ and $n \geq 1$, put

$$W_{d,n} = \frac{n!}{(d!)^{\lfloor n/d \rfloor}}.$$

Writing $n = kd + r$, $0 \leq r < d$, shows that $W_{d,n}$ is $r!$ times the multinomial coefficient $n!/((d!)^k r!)$. In particular it is an integer, and $n! = (d!)^k W_{d,n}$ gives $W_{d,n} \equiv n! \pmod{d! - 1}$. For example, $W_{3,4} = 4$ and $(4! - W_{3,4})/(3! - 1) = 4$. The exponent $\lfloor n/d \rfloor$ increases precisely at multiples of d , which will determine the nonzero weighted differences.

For a finitely supported integer vector $\lambda = (\lambda_n)_{n \geq 1}$, define the integer linear forms

$$M(\lambda) = \sum_n \lambda_n n!, \quad V_d(\lambda) = \sum_n \lambda_n W_{d,n},$$

and let

$$\mathcal{R}(\lambda) = \sum_{d \geq 2} \frac{V_d(\lambda)}{d! - 1}.$$

Here n indexes the chosen coefficients, whereas d indexes the summands of the remainder. If λ is supported on $n \leq N$, with $N \geq 2$, then $d > N$ makes every floor exponent

zero. Hence $V_d = M$ throughout this tail, and the series converges absolutely. Only the first $N - 1$ terms can differ from those of MS ; the weight congruence gives

$$\mathcal{R}(\lambda) - M(\lambda)S = \sum_{d=2}^N \frac{V_d(\lambda) - M(\lambda)}{d! - 1} \in \mathbb{Z}. \quad (1)$$

We now solve $V_2 = \dots = V_D = 0$ in integer vectors, allowing either sign of M . Write e_n for the unit vector at index n . The auxiliary coordinate at index one will make the basis triangular. It occurs only in the coefficient vector: the series still starts at $d = 2$, so no term $1/(1! - 1)$ is introduced. We impose $\lambda_1 = 0$ in Section 3.

The identity $n(n-1)! - n! = 0$ suggests starting with $T_n = ne_{n-1} - e_n$. At $n = 2$, the vector $U_2 = 2e_1 - e_2$ has $V_2 = 1$ and every other weighted sum zero. At $n = 4$, the difference $T_4 = 4e_3 - e_4$ has two nonzero weighted sums. One correction suffices:

Vector	M	V_2	V_3	V_4
T_4	0	6	0	23
$6U_2$	0	6	0	0
$T_4 - 6U_2$	0	0	0	23

For $d > 4$ these weighted sums equal $M = 0$. Thus $U_4 = T_4 - 6U_2$ isolates $V_4 = 4! - 1$. In general, the unwanted sums occur at proper divisors of n , whose correcting vectors are already available in an inductive construction.

Theorem 2.1 (an integer basis with prescribed weighted sums). *Set*

$$T_n = ne_{n-1} - e_n, \quad U_n = T_n - \sum_{\substack{d|n \\ 2 \leq d < n}} W_{d,n} U_d \quad (n \geq 2).$$

Then

$$M(U_n) = 0, \quad V_d(U_n) = (d! - 1)\mathbf{1}_{d=n}.$$

The vectors $e_1, U_2, U_3, \dots$ form an integral basis. Every finite vector has the unique finite expansion

$$\lambda = M(\lambda)e_1 + \sum_{d \geq 2} \frac{V_d(\lambda) - M(\lambda)}{d! - 1} U_d. \quad (2)$$

Proof. We have $M(T_n) = 0$. If $d \nmid n$, the floor exponents at $n-1$ and n agree, so $nW_{d,n-1} = W_{d,n}$. If $d \mid n$, the exponent increases by one, so $nW_{d,n-1} = d!W_{d,n}$. Hence

$$V_d(T_n) = (d! - 1)W_{d,n}\mathbf{1}_{d|n}.$$

Suppose the asserted identities hold below n . For each proper divisor d , subtracting $W_{d,n}U_d$ cancels the sum at d and changes none of the others. The sum at n remains $n! - 1$, since $W_{n,n} = 1$. This proves the identities by induction.

For completeness, restrict to indices $1, \dots, N$. The columns $e_1, T_2, \dots, T_N$ form a triangular integer matrix with diagonal $1, -1, \dots, -1$. Replacing each T_n by U_n subtracts an integer combination of earlier columns, so the second change of basis is triangular with

diagonal one. Both matrices have integer inverses: back-substitution divides only by 1 or -1 . Hence $e_1, U_2, \dots, U_N$ is a basis over $\mathbb{Z}$.

In the resulting expansion, applying M gives the coefficient of e_1 . If the coefficient of U_d is c_d , applying V_d gives $V_d = M + (d! - 1)c_d$, which proves (2). For $d > N$ that coefficient is zero because $V_d = M$. $\square$

The U_d -coordinates in (2) are exactly the integer summands in (1). Their sum is therefore k in $\mathcal{R} = MS + k$. The same coordinates describe both the vector and its linear form.

For a prime $p \geq 3$ the divisor sum is empty, and $U_p = pe_{p-1} - e_p$. At a composite index the correction terms remove the extra weighted sums. For example,

$$U_9 = 9e_8 - e_9 - 5040e_2 + 1680e_3. \quad (3)$$

Since $M(U_n) = 0$ and $\mathcal{R}(U_n) = 1$, adding an integer multiple of U_n changes only V_n among the weighted sums and changes the remainder by an integer. It therefore preserves both M and the fractional part of the remainder.

3 Possible values of M

Fix $D \geq 2$. We first solve $V_2 = \dots = V_D = 0$ with index one allowed, and then impose the support restriction $\lambda_1 = 0$. The weight congruence, or the coordinates in (2), gives

$$V_d(\lambda) \equiv M(\lambda) \pmod{d! - 1}. \quad (4)$$

It follows that M must be divisible by

$$L_D = \text{lcm}_{2 \leq d \leq D}(d! - 1).$$

This necessary condition is sufficient while index one is allowed: L_De_1 has every weighted sum equal to L_D , and subtracting $L_D/(d! - 1)$ copies of U_d sets the d th sum to zero. Thus put

$$K_D = L_De_1 - \sum_{d=2}^D \frac{L_D}{d! - 1} U_d.$$

Uniqueness of the coordinates now gives the complete solution

$$V_2 = \dots = V_D = 0 \iff \lambda = tK_D + \sum_{n>D} z_n U_n, \quad M = tL_D, \quad (5)$$

with $t \in \mathbb{Z}$ and finitely many nonzero integers z_n . The lower coordinates are fixed by t ; all higher coordinates remain free. These free coordinates help impose the support restriction, but they change the remainder only by an integer. Indeed, $\mathcal{R}(e_1) = S$ and $\mathcal{R}(U_n) = 1$ give

$$\mathcal{R}\left(tK_D + \sum_{n>D} z_n U_n\right) = tL_D(S - H_D) + \sum_{n>D} z_n, \quad H_D = \sum_{d=2}^D \frac{1}{d! - 1}. \quad (6)$$

Since $M = tL_D$ clears the denominators of H_D , the integer in $\mathcal{R} = MS + k$ is $k = -MH_D + \sum_{n>D} z_n$. Thus the sum of the free coordinates is the offset from $M(S - H_D)$, not from MS .

It remains to remove index one. Write a_D for its coefficient in K_D , and u_n for its coefficient in U_n . All other coordinates already lie in the permitted range $n \geq 2$, so the only remaining equation is

$$ta_D + \sum_{n>D} z_n u_n = 0. \quad (7)$$

The higher corrections can change this coefficient by exactly the integer multiples of $g_D = \gcd\{u_n : n > D\}$. Thus the equation is soluble precisely when $g_D \mid ta_D$. For example, at depth four we shall find $a_4 = -55$ and $g_4 = 60$. Removing index one then requires $60 \mid 55t$, or $12 \mid t$; hence $M = 115t$ must be a multiple of 1380.

In general, the possible values of M form the ideal

$$\mu_D \mathbb{Z}, \quad \mu_D = L_D \frac{g_D}{\gcd(g_D, a_D)}. \quad (8)$$

Indeed, division by $\gcd(g_D, a_D)$ leaves two coprime integers, so $g_D \mid ta_D$ holds exactly when $g_D / \gcd(g_D, a_D) \mid t$. Bézout's identity supplies the required finite combination of the u_n . The next theorem proves $g_D > 0$ and bounds the indices needed for this combination. A vector attaining the least positive value μ_D is primitive, since dividing its coefficients by a common factor would give a smaller positive value of M .

3.1 A finite gcd calculation

The recursion for the first coordinate is

$$u_2 = 2, \quad u_n = - \sum_{\substack{d \mid n \\ 2 \leq d < n}} W_{d,n} u_d \quad (n > 2).$$

Induction gives $u_n = 0$ at odd indices. At twice a prime it gives $u_{2p} = -(2p)!/2^{p-1} \neq 0$, including $p = 2$, so $g_D > 0$. This nonzero coefficient also tells us how to make the calculation finite. Choose a prime ℓ with $D/2 < \ell \leq D$. Any gcd computed on an initial interval of the tail containing 2ℓ must divide $(2\ell)!$. We use that factorial to control the later terms of the recurrence.

For a proper divisor $d > D$, divisibility by the finite gcd comes from the earlier coefficient u_d . For $d \leq D$, we instead make the weight $W_{d,n}$ divisible by $(2\ell)!$. The proof obtains this from $n/d \geq 2\ell$; the cutoff below is chosen to ensure that inequality for every small divisor.

Theorem 3.1 (a finite formula for the gcd). *Choose a prime ℓ with $D/2 < \ell \leq D$ and put $H = D(2\ell - 1)$. Then*

$$g_D = \gcd(u_{D+1}, \dots, u_H), \quad H < 2D^2.$$

Proof. Let $g = \gcd(u_{D+1}, \dots, u_H)$. The index 2ℓ belongs to this interval, and $u_{2\ell} = -(2\ell)!/2^{\ell-1} \neq 0$. Consequently $g > 0$ and $g \mid (2\ell)!$.

We prove $g \mid u_n$ for every $n > D$ by strong induction. This is true up to H by definition. For $n > H$, consider a proper divisor $d \geq 2$ in the recurrence for u_n . If $d > D$, the induction hypothesis gives $g \mid u_d$. If $d \leq D$, put $k = n/d$. The choice $H = D(2\ell - 1)$ ensures $k \geq 2\ell$. Moreover,

$$\frac{W_{d,n}}{k!} = \frac{n!}{(d!)^k k!} \in \mathbb{Z} :$$

this quotient counts partitions of n labelled objects into k unordered blocks of size d . Hence $(2\ell)! \mid k! \mid W_{d,n}$, so $g \mid W_{d,n}$ in the small-divisor case as well. Every product $W_{d,n}u_d$ in the recurrence is therefore divisible by g .

Thus g divides the whole tail. Conversely, every common divisor of the tail divides its finite subfamily $u_{D+1}, \dots, u_H$. The two gcds are equal, and $H \leq D(2D-1) < 2D^2$. $\square$

Bertrand's postulate supplies the prime ℓ when $D \geq 3$; for $D = 2$ take $\ell = 2$. Merely including the nonzero coefficient at 2ℓ is not enough: the cutoff must also control every later small-divisor weight. A truncation at $2D$ is not justified by this argument.

The finite gcd also gives a construction. Compute integers b_n with $\sum_{n=D+1}^H b_n u_n = g_D$ by the extended Euclidean algorithm, and put $h = \gcd(g_D, a_D)$. Taking

$$t = g_D/h, \quad z_n = -(a_D/h)b_n \quad (D < n \leq H)$$

makes $ta_D + \sum z_n u_n = 0$. Substitution in (5) therefore produces a vector with $M = \mu_D$. This gives an attaining vector as well as the minimum. Minimising its support or its coefficient norm is a separate problem.

4 An example attaining the minimum

At depth four the recursion gives

$$L_4 = 115, \quad K_4 = -55e_1 + 16e_2 + 3e_3 + 5e_4, \quad a_4 = -55.$$

Take $\ell = 3$ in Theorem 3.1, so that $H = 20$. The coefficients in this finite interval have gcd 60. There is a useful short proof: $u_6 = -180$, $u_8 = -4200$ and $23u_6 - u_8 = 60$, while

$$\text{lcm}(1, \dots, n) \mid u_n \tag{9}$$

gives the reverse divisibility for every $n > 4$. To prove (9), put $\Lambda_n = \text{lcm}(1, \dots, n)$. For a prime r , write v_r for its valuation. If $d \mid n$, Legendre's formula gives

$$v_r(W_{d,n}) = \sum_{j \geq 1} \left(\left\lfloor \frac{n}{r^j} \right\rfloor - \frac{n}{d} \left\lfloor \frac{d}{r^j} \right\rfloor \right) \geq v_r(\Lambda_n) - v_r(\Lambda_d).$$

Each summand is nonnegative, and each power $d < r^j \leq n$ contributes at least one. Hence $\Lambda_n \mid W_{d,n}\Lambda_d$, and induction in the recurrence, starting from $u_2 = 2$, proves the assertion. Formula (8) now gives

$$\mu_4 = 115 \frac{60}{\gcd(60, 55)} = 1380.$$

To attain $M = 1380$, take $t = 12$ in (7). The choice $z_6 = -27, z_8 = 1$ works because

$$12(-55) - 27(-180) - 4200 = 0.$$

The vector $12K_4 - 27U_6 + U_8$ therefore has no index-one coefficient and cancels V_2, V_3, V_4 . To remove its coefficients at indices four and seven, add $-26U_5 + 8U_7$. These corrections still cancel through four and do not restore index one, since $u_5 = u_7 = 0$. Thus

$$\lambda = 12K_4 - 26U_5 - 27U_6 + 8U_7 + U_8,$$

which expands to

$$\lambda = 1482e_2 - 784e_3 - 136e_5 + 83e_6 - e_8, \quad (M, V_2, V_3, V_4) = (1380, 0, 0, 0). \quad (10)$$

Table 1 verifies all four sums directly.

n	λ_n	$n!$	$W_{2,n}$	$W_{3,n}$	$W_{4,n}$
2	1482	2	1	2	2
3	-784	6	3	1	6
5	-136	120	30	20	5
6	83	720	90	20	30
8	-1	40320	2520	1120	70
Weighted totals		1380	0	0	0

Table 1: The depth-four vector and its checks. To obtain the last row, multiply each of the final four columns by λ_n before summing.

The coordinates give $M = 1380$, the three cancellations, and the offset $-26 - 27 + 8 + 1 = -44$ in (6). Since $H_4 = 143/115$, the original linear form is

$$\mathcal{R}(\lambda) = 1380(S - H_4) - 44 = 1380S - 1760.$$

For the remainder estimate, the inequality $(n+1)! - 1 > (n+1)(n! - 1)$ gives the geometric bound

$$0 < S - H_N < \frac{N+2}{(N+1)((N+1)! - 1)} < \frac{2}{(N+1)! - 1} \quad (N \geq 2). \quad (11)$$

The majorant starts with $1/((N+1)! - 1)$ and has ratio $1/(N+2)$.

Exact rational arithmetic gives

$$-31 + \frac{4}{5} < 1380 \sum_{d=5}^8 \frac{1}{d! - 1} - 44 < -31 + \frac{5}{6}, \quad \frac{2 \cdot 1380}{9! - 1} < \frac{1}{100}.$$

The positive remaining tail is less than $2/(9! - 1)$, so $-31 < \mathcal{R}(\lambda) < -30$. If $S = a/q$ with $q \mid 1380$, (1) would make this remainder an integer. Every divisor of 1380 is therefore excluded.

The companion gives the [alternative dual congruence, support and coefficient-norm refinements](#). These minimise different quantities. The minimum of M follows from (8) and does not require minimal support.

5 Nonintegrality of the remainder

Fix a vector with $V_2 = \dots = V_D = 0$ and $M > 0$. We may choose the cutoff $N \geq D$ anywhere at or beyond its support. Separate the finite signed sum from the positive tail:

$$A_N = \sum_{d=D+1}^N \frac{V_d(\lambda)}{d! - 1}, \quad \mathcal{R}(\lambda) = A_N + M(S - H_N).$$

The remainder lies above A_N . To place it below the next integer, it is enough that the omitted tail be shorter than the gap $\lfloor A_N \rfloor + 1 - A_N$. By (11), a sufficient condition is

$$\frac{2M}{(N+1)! - 1} < \lfloor A_N \rfloor + 1 - A_N. \quad (12)$$

Indeed, the positive tail and this comparison give $\lfloor A_N \rfloor < \mathcal{R}(\lambda) < \lfloor A_N \rfloor + 1$, excluding every denominator dividing M . Failure of this sufficient comparison is inconclusive. Either sign of A_N is allowed: its strict gap is one when A_N is integral and can be arbitrarily small when A_N approaches an integer from below.

First keep the vector fixed and increase only the cutoff. The vector $-6e_2 + e_4$ has $M = 12$. At $N = 4$, its finite part $A_4 = -239/115$ lies just below -2 , and its gap $9/115$ is smaller than the bound $24/119$. At $N = 5$, the finite part $A_5 = -27061/13685$ has crossed -2 : the next integer is now -1 , and the gap $13376/13685$ exceeds $24/719$. More generally, for a fixed vector with $M > 0$, the quantities A_N increase to $\mathcal{R}$. If this limit is nonintegral, their gaps to the next integer tend to a positive number, so the test eventually succeeds. If the limit is integral, the eventual gap equals the omitted tail and the test fails. Extending the cutoff therefore eventually detects a nonintegral remainder, but leaves both that remainder and the set of divisors of M unchanged.

Changing coefficients while preserving M also leaves integrality unchanged. By (1), the remainders of two such vectors differ by an integer. Adding multiples of the U_n can simplify the support and weighted sums, as in the example; it cannot turn an integral remainder into a nonintegral one.

To exclude every possible denominator, we must instead choose vectors whose values of M have the required divisibility. There are primitive solutions with $0 < M < N!$ for which M is eventually divisible by any prescribed q . The [progression construction in the companion](#) gives such a family by turning $V_d = 0$ into roots of a polynomial. At its support endpoint the omitted tail is less than $2/N$, but the gap on the right of (12) may shrink too. To prove irrationality by this test, for each $q > 0$ we need one vector with $q \mid M$ and a cutoff for which (12) holds. Divisibility and a small tail do not by themselves establish that comparison. No minimal-support assertion is made for this family.

The companion treats the [equivalent carry condition](#) and the [complementary-denominator tail inequality](#). It also records two larger finite computations: a non-unit carry at $m = 300000$ excludes every $q \mid 299999!$, and a continued-fraction enclosure excludes every $q < 2^{39990}$. Both computations were performed outside Lean; their [algorithms](#), [exact outputs](#) and [scope](#) are given there. The factorial-divisibility exclusion does not exclude all denominators with only small prime factors, whose exponents may be larger. Neither finite restriction establishes the required condition at arbitrarily large indices.

6 Verification

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The [verification record](#) gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded.

The integral-basis theorem and finite gcd formula have recorded exact Lean bindings. The [companion source concordance](#) identifies their declarations and the theorem on the possible values of M . Its large finite computations are separate from these kernel proofs.

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References

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- [4] Jaroslav Hančl and Robert Tijdeman. [On the irrationality of factorial series](#). *Acta Arithmetica* **118** (4) (2005), 383–401.

Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 2.1	Lean · Comparator	Theorem 3.1	Lean · Comparator